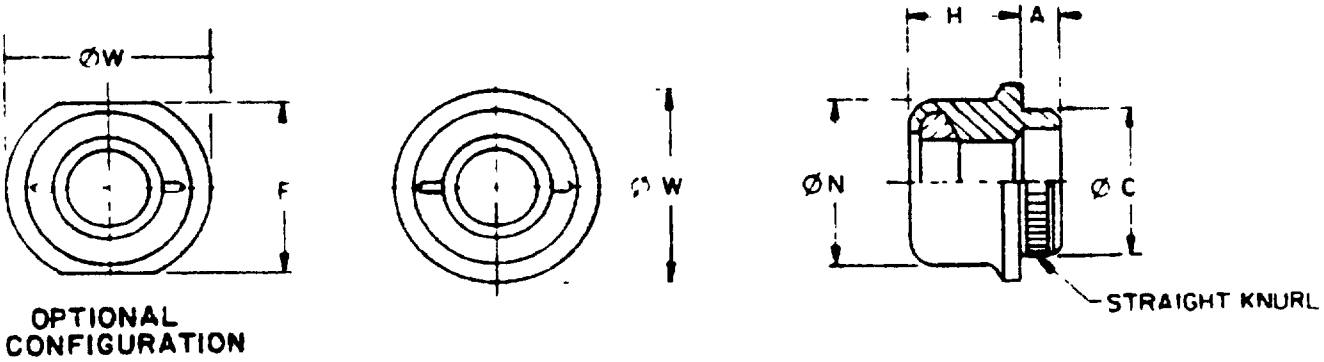
MILITARY SPECIFICATION SHEET

NUT, SELF-LOCKING, CLINCH, (SWAGE-CLINCHING, KNURLED SHANK, MINIATURE, 250°F)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nuts described herein shall consist of this specification and the latest issue of MIL-N-45938.



NOTES:

1. All dimensions are in inches.

FIGURE 1. NUT, SELF-LOCKING, CLINCH, (SWAGE-CLINCHING, KNURLED SHANK, MINIATURE, 250°F).

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TABLE I. DASH NUMBERS AND DIMENSIONS.

DASH NO		THREAD SIZE	A SHANK LENGTH ±.003	SHEET THICKNESS	HOLE SIZE IN SHEET ±.001	ØC ±.002	H ±.010	ØN +.000 -.010	ØW ±.005	F ±.005
CARBON OR ALLOY STEEL[1]	CRES									
1	1C	.086-56UNJC-3B	.040	.030 TO .045	.125	.129	.075	.150	.172	.156
2	2C		.060	.046 AND UP						
3	3C	.112-40UNJC-3B	.040	.030 TO .045	.156	.160	.090	.182	.203	.187
4	4C		.060	.046 AND UP						
5	5C	.138-32UNJC-3B	.040	.030 TO .045	.188	.192	.130	.242	.281	.238
6	6C		.060	.046 AND UP						
7	7C	.164-32UNJC-3B	.060	.046 AND UP	.219	.223	.160	.268	.312	.281
8	8C	.190-32UNJF-3B	.060	.046 AND UP	.250	.254	.179	.287	.344	.328

1/ For Size-Material Combinations: See Requirement 1.

MIL-H-8933/56

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REQUIREMENTS:

1. Material. Carbon steel, Grade 1012 (UNS G10120) in accordance with ASTM A510, or Grade 1018 (UNS G10180) with lead for free machining, in accordance with ASTM A108 for thread sizes .086-56UNJC-3B thru .112-40UNJC-3B.
Alloy steel, Grade 4130 (UNS G41300) in accordance with ASTM A322 or ASTM A331 shall be used for thread sizes .138-32UNJC-3B and larger.
Corrosion resistant steel, Type 303 (UNS S30300) in accordance with ASTM A581 or ASTM A582, shall be used for all thread sizes. The Locking Insert shall be nylon.

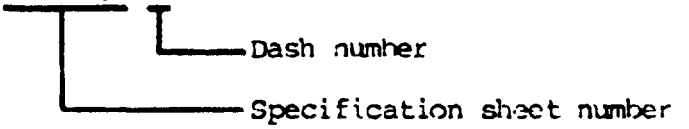
2. Protective coating and surface treatment. Carbon Steel, Grades 1012 (UNS G10120) and 1018 (UNS G10180), shall be cadmium plated. Corrosion resistant steel, Type 303 (UNS S30300), shall be cleaned, descaled and passivated.

3. Threads. Threads shall be in accordance with MIL-S-8879.

4. Push-out and torque-out. Push-out and torque-out values shall be as specified in Table II when properly installed in low carbon steel sheet.

5. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I.

Example: M45938/5-1



M45938/5-1 indicates - Nut, Self-locking, Clinch, (Swage-Clinching, Knurled Shank, Miniature, 250°F), Carbon steel, .086-56UNJC-3B, .040 Shank Length.

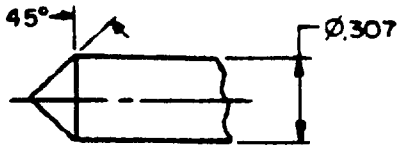
TABLE II. PERFORMANCE AND INSTALLATION DATA.

THREAD SIZE	PUSH OUT LB MIN	TORQUE-OUT IN LB MIN	APPROX INSTALLATION FORCE LB
.086-56UNJC-3B	20	5	300
.112-40UNJC-3B	40	5	600
.138-32UNJC-3B	60	8	750
.164-32UNJC-3B	80	12	1000
.190-32UNJF-3B	100	15	1500

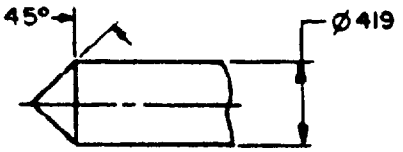
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RECOMMENDED INSTALLATION PROCEDURE

a. Drill or punch hole.
b. Press knurled shank of nut into hole using a dolly as shown below until the shoulder of the nut contacts the structure for a full 360°.
c. Using the dolly to support the nut head, flare the shank with one of the flaring tools shown below. Do not exceed the installation forces specified in Table II.
d. Install the screw or bolt from the shank end of the nut.



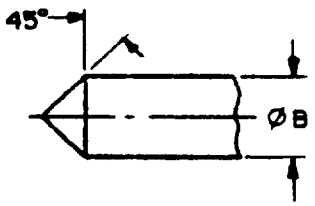
FOR THREAD SIZES 086,.112 AND .138



FOR THREAD SIZES .164 AND .190

FLUSH FLARING TOOL

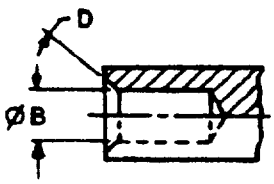
Use when sheet thickness is equal to or less than shank length (A). See Table I.



NUT SIZE	Ø B +.000 -.002
.086	.120
.112	.151
.138	.183
.164	.214
.190	.245

INTERNAL FLARING TOOL

Use when sheet thickness is greater than shank length (A). See Table I.



NUT SIZE	Ø B +.003 -.000	D ±1°
.086	.151	35°
.112	.183	
.138	.243	21°
.164	.269	
.190	.288	

DOLLY

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Army - AR
Navy - CG

Review activities:
Army - AV, GL
DLA - IS
NSA - NS

User activities:
Army - AT, ER
Navy - AS, MC, SH

Agent:
DLA - IS

Preparing activity:
Army - AR

(Project 5310-1340)

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INCH-POUND
MIL-N-45938/5A
AMENDMENT 2
1 February 1990
SUPERSEDING
AMENDMENT 1
31 July 1986

MILITARY SPECIFICATION SHEET
NUT, SELF LOCKING, CLINCH
(SWAGE-CLINCHING, KNURLED SHANK, MINATURE, 250°F)

This amendment forms a part of MIL-N-45938/5A, dated 6 November 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 2

Table I, Column F Delete " ± 005" and substitute "± 015".

*Table I, Column ØN Delete "+ 000" and substitute "MAX"
- 010

NOTE: The margins of this amendment are marked with an asterick or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations Bidders and contractors are cautioned to evalutate the requirements of this document based relationship to the last previous amendment

Custodians:
Army - AR
Navy - OS

Preparing activity:
Army - AR

Review activities
Army - AV, GL
DLA - IS
NSA - NS

(Project 5310-1761)

User activities:
Army - AT, ER
Navy - AS, MC, SH

AMSC N/A FSC 5310
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